

Here is what Docling delivers today:

- Converts PDF documents to JSON or Markdown format, stable and lightning fast
- Understands detailed page layout, reading order, locates figures and recovers table structures
- Extracts metadata from the document, such as title, authors, references and language
- Optionally applies OCR, e.g. for scanned PDFs
- Can be configured to be optimal for batch-mode (i.e high throughput, low time-to-solution) or interactive mode (compromise on efficiency, low time-to-solution)
- Can leverage different accelerators (GPU, MPS, etc).

2 Getting Started

To use Docling, you can simply install the [docling](#) package from PyPI. Documentation and examples are available in our [GitHub](#) repository at [github.com/DS4SD/docling](#). All required model assets¹ are downloaded to a local huggingface datasets cache on first use, unless you choose to pre-install the model assets in advance.

Docling provides an easy code interface to convert PDF documents from file system, URLs or binary streams, and retrieve the output in either JSON or Markdown format. For convenience, separate methods are offered to convert single documents or batches of documents. A basic usage example is illustrated below. Further examples are available in the Docling code repository.

```
from docling.document_converter import DocumentConverter

source = "https://arxiv.org/pdf/2206.01062" # PDF path or URL
converter = DocumentConverter()
result = converter.convert_single(source)
print(result.render_as_markdown()) # output: "## DocLayNet: A Large
    Human-Annotated Dataset for Document-Layout Analysis [...]"
```

Optionally, you can configure custom pipeline features and runtime options, such as turning on or off features (e.g. OCR, table structure recognition), enforcing limits on the input document size, and defining the budget of CPU threads. Advanced usage examples and options are documented in the README file. Docling also provides a *Dockerfile* to demonstrate how to install and run it inside a container.

3 Processing pipeline

Docling implements a linear pipeline of operations, which execute sequentially on each given document (see Fig. 1). Each document is first parsed by a PDF backend, which retrieves the programmatic text tokens, consisting of string content and its coordinates on the page, and also renders a bitmap image of each page to support downstream operations. Then, the standard model pipeline applies a sequence of AI models independently on every page in the document to extract features and content, such as layout and table structures. Finally, the results from all pages are aggregated and passed through a post-processing stage, which augments metadata, detects the document language, infers reading-order and eventually assembles a typed document object which can be serialized to JSON or Markdown.

3.1 PDF backends

Two basic requirements to process PDF documents in our pipeline are a) to retrieve all text content and their geometric coordinates on each page and b) to render the visual representation of each page as it would appear in a PDF viewer. Both these requirements are encapsulated in Docling's PDF backend interface. While there are several open-source PDF parsing libraries available for python, we faced major obstacles with all of them for different reasons, among which were restrictive

¹see [huggingface.co/ds4sd/docling-models/](#)

murder	n. 谋杀	pollute	vi. 污染
native	a. 本土的	powder	n. 粉末, 火药
naturally	ad. 自然地	power	n. 力量
nerous	a. 紧张的	powerful	a. 强有力的
nerousness	n. 神经过敏	practice	v. 练习, 实践
nomads	n. 游牧部落	prefer	v. 更喜欢
nut	n. 坚果	president	n. 总统, 校长
nutritious	a. 有营养的	pressure	v. 向...施压力 n. 压力
obstacle	n. 障碍	private	a. 私人的
operation	n. 操作; 经营; 手术	produce	vt. 生产
orbit	n. 运行轨道 vt. 环绕	producer	n. 生产者
organize	v. 组织	production	n. 产品, 产量
ornithologist	n. 鸟类学者	progress	n. 进步
outgoing	adj. 友善的, 将离去的	prompt	n. 提示
overacted	adj. 行为夸张的	protect	vt. 保护
partnership	n. 合作	protein	n. 蛋白质
patient	a. 耐心的 n. 病人	prove	vb. 证明 vi. 结果是
peacefully	ad. 和平地	psychologist	n. 心理学家
peanut	n. 花生	punish	v. 惩罚
perfect	vt. 改善 a. 完美的	purpose	n. 目的
perform	vt. 执行, 演出	radio	v. 用无线电发射
phone	n. 电话 vt. 打电话	rainforest	n. 雨林
photography	n. 摄影	rate	n. 比率, 速度 vt. 评价
physically	ad. 体格上	raw	a. 生的, 未加工的
picture	v. 描绘	reference	n. 证明人
poison	n. 毒药 vt. 毒害	refuse	v. 拒绝